

3. (a) A student wanted to investigate the concentration of casein in milk powder. Casein is an insoluble protein found in milk. When milk powder is mixed with water it forms an opaque (cloudy) suspension. The enzyme trypsin breaks down the casein into soluble peptides, forming a transparent solution.

To begin with the student needed to construct a calibration curve from known concentrations of milk powder suspensions. He was provided with the following:

- 10 cm<sup>3</sup> suspensions made from milk powder at concentrations of 1%, 2%, 3%, 4% and 5% in separate test tubes with a buffer
- 1% trypsin solution

The milk suspensions and trypsin solution were placed separately in a thermostatically controlled water bath at 30 °C. Once both the milk suspensions and trypsin solution were at the correct temperature, he transferred 2 cm<sup>3</sup> of trypsin to each of the suspensions of milk powder and timed how long it took for the suspensions to become transparent.

- (i) Using your knowledge of enzymes, explain why:

I. it was necessary for a buffer to be added to the solutions;

[2]

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II. all test tubes were placed in a thermostatically controlled water bath.

[2]

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The table below shows the results.

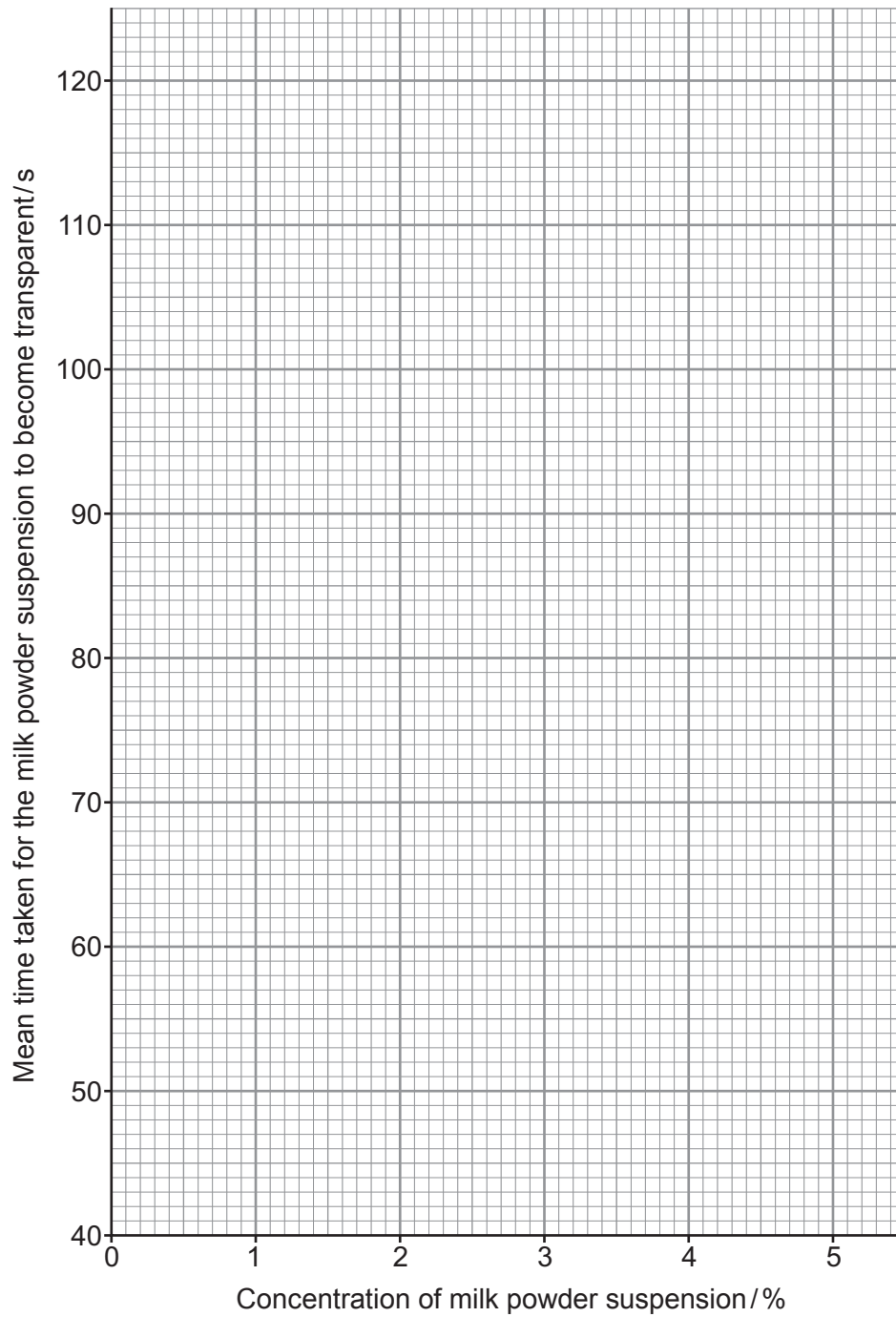
| Concentration of milk powder suspension / % | Time taken for milk powder suspension to become transparent / s |         |         |         |         |       |
|---------------------------------------------|-----------------------------------------------------------------|---------|---------|---------|---------|-------|
|                                             | Trial 1                                                         | Trial 2 | Trial 3 | Trial 4 | Trial 5 | mean  |
| 1                                           | 49                                                              | 51      | 55      | 52      | 53      | 52.0  |
| 2                                           | 66                                                              | 64      | 62      | 61      | 63      | 63.2  |
| 3                                           | 88                                                              | 77      | 73      | 81      | 84      | 80.6  |
| 4                                           | 91                                                              | 87      | 93      | 86      | 99      | ..... |
| 5                                           | 96                                                              | 109     | 108     | 111     | 105     | 105.8 |

- (ii) Calculate the mean time taken for the 4% milk powder suspension to become transparent. **Write your answer in the table.**

[1]



- (iii) Plot the mean data on the graph paper provided. **Range bars should also be added to the graph.** [4]



- (b) The student was then provided with two different suspensions of milk powder, **X** and **Y**, of unknown casein concentration. He carried out the same procedure and obtained the following mean results: **X** took 61 seconds to clear and **Y** took 97 seconds to clear.

- (i) Estimate the percentage concentration of casein in these two suspensions. [1]

**X:** ..... **Y:** .....

- (ii) The student had more confidence in the estimate for suspension **X** than the estimate for suspension **Y**. Use the graph to explain the reason for this statement. [2]

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- (iii) Suggest **two** sources of inaccuracy in this experiment. [2]

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